

OPTION B INSTITUTE FAMILY

M00 - COMPUTER, SETUP & LEARNING FOUNDATIONS

Standalone Zero-to-Job-Ready Data Engineer 2026

BEGINNER-FIRST MASTER REPAIR RC4

8 canonical lessons | full 16-stage teacher contract | protected attempt-first review

This learner book is rebuilt from the frozen canonical technical material. It is not a front-matter patch: every lesson is re-laid out into the shared institute teaching grammar while preserving the course competency, source route, guided work, independent proof and validation boundary.

This book is your teacher

Use the book in order. External videos and official docs are visual companions, not hidden prerequisites. Watch or read only the assigned scope, then close it before the independent task.

The 16-stage lesson loop

Why it matters -> What you learn -> Prerequisite -> Picture it -> New words -> Useful source -> Follow once -> Expected result -> Why it worked -> Beginner mistakes -> Recovery -> Independent task -> Validate -> Teach it back -> Protected review + changed retry -> Evidence/mastery + exact next route.

Attempt-first protection

Never open a protected lesson review before saving a genuine independent attempt. If repair is needed, compare only the weak idea, close the review, solve a changed retry, then validate again. A solution seen during review is repair evidence, not mastery.

What counts as mastery

A reproducible artifact or explanation, an independent validation control, preserved failure/recovery evidence, a changed transfer task completed without the worked answer open, and an own-words explanation. Percent-complete is not a substitute.

How to use your devices

Phone: assigned video/reference, reading, revision. Laptop/PC: every real command, query, file, workbook, code task, validation and evidence artifact. Install only what the current lesson asks for.

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Prerequisite / tool map

Become safe and independent with files, folders, software, VS Code, terminal use and study evidence.

Item	Rule
Entry prerequisite	Absolute beginner entry; no prior technical background required.
Primary device	PC/laptop for hands-on work; phone is support only.
External sources	Optional/supporting unless the lesson explicitly marks an official current reference.
Evidence	Module practical + fresh retest + saved evidence; no critical unresolved failure.

Practice / review / Gate route

Lesson teaching -> follow once -> independent attempt -> validate -> save evidence. Only after the genuine attempt may protected review open. If weak: targeted repair -> close review -> changed retry -> revalidate. Finish the module practical/retest route before any downstream Gate that depends on this module.

DE00-01 - L01 - Files, Folders, Drives and Paths

1. Why it matters

Later tools fail when the learner cannot distinguish file identity from file location.

2. What you are learning

Build and explain the competency: L01 - Files, Folders, Drives and Paths.

3. Prerequisite / starting state

Starting state: no prior technical knowledge is assumed beyond the module Start Here instructions.

4. Picture it in your head

A file name is like a person name; a path is the full address that tells the computer where the item lives.

5. New words before we use them

A file name is like a person name; a path is the full address that tells the computer where the item lives.

6. Useful video / source

Source: Microsoft Support — File Explorer in Windows

Assigned scope: Files/folders, navigation, copy/move and organization

How to use it: Use this current Windows guide as a visual companion to the Teaching Book. Do the course folder/path task yourself; no handwritten notes.

Note: Official Windows guide; Teaching Book remains authoritative for the course evidence-folder convention.

[Open source](#)

7. Follow with me once

Create Documents\DE_COURSE_PRACTICE\files_lab, add location_test.txt, reopen File Explorer and locate it without Recent. Create Documents\DE_COURSE_PRACTICE\files_lab, add location_test.txt, reopen File Explorer and locate it without Recent.

8. Expected result / what you should see

One stable course folder, one files_lab folder and one location_test.txt that remains after File Explorer is closed.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Looking in Downloads when the file is in Documents.
- Creating project_final_final folders instead of one stable workspace.
- Using Recent instead of knowing the actual location.

11. If you get stuck

Stop at the first point where your result diverges. Restate the intended input/state and expected output, inspect the smallest relevant step, reproduce only that step, and keep the failed attempt as evidence. Open protected review only after a genuine attempt has been saved.

12. Small independent task

Inside files_lab create practice/my_address_test.txt. Close File Explorer, reopen from Documents, find the file without Recent and write its full path inside the file. Evidence: Screenshot or note containing the exact folder path and the full file path.

13. Validate

One stable course folder, one files_lab folder and one location_test.txt that remains after File Explorer is closed. Save a count, schema/key check, reconciliation, observed state, test, or other independent control appropriate to the lesson.

14. Teach it back

- File vs folder?
- What is missing if I only say orders.csv?
- What does a path tell you?

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE00-02

DE00-02 - L02 - File Extensions and Common Data Files

1. Why it matters

Data engineering tools choose parsers from the real format, not from the icon shown in Windows.

2. What you are learning

Build and explain the competency: L02 - File Extensions and Common Data Files.

3. Prerequisite / starting state

Prerequisite: DE00-01 mastered with saved evidence.

4. Picture it in your head

The extension describes the file format. The app icon only shows a default way Windows may open it.

5. New words before we use them

The extension describes the file format. The app icon only shows a default way Windows may open it.

6. Useful video / source

Source: Microsoft Support — Common file name extensions in Windows

Assigned scope: Windows 10/11: what extensions mean, examples, showing extensions, and why renaming does not convert a format

How to use it: Use the current Microsoft guide beside the Teaching Book, then identify .csv/.json/.xlsx/.pdf/.py/.sql/.zip files on your own PC. No handwritten notes.

Note: Official current Windows reference. The course lab remains the practical teacher for data-file handling.

[Open source](#)

7. Follow with me once

Turn on File name extensions. Open sample.csv first in a text editor and then in Excel if available; observe that it remains CSV. Turn on File name extensions. Open sample.csv first in a text editor and then in Excel if available; observe that it remains CSV.

8. Expected result / what you should see

Extensions are visible and the same CSV can be inspected as comma-separated text and as rows/columns.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Renaming .csv to .xlsx and assuming conversion happened.
- Hiding extensions and creating report.csv.csv.
- Opening code files in rich-text software that can damage plain text.

11. If you get stuck

Stop at the first point where your result diverges. Restate the intended input/state and expected output, inspect the smallest relevant step, reproduce only that step, and keep the failed attempt as evidence. Open protected review only after a genuine attempt has been saved.

12. Small independent task

Choose four bundled files, predict each format from its extension before opening, then verify with a suitable app. Evidence: A four-row note: filename, predicted format, app used, what you observed.

13. Validate

Extensions are visible and the same CSV can be inspected as comma-separated text and as rows/columns. Save a count, schema/key check, reconciliation, observed state, test, or other independent control appropriate to the lesson.

14. Teach it back

- Which part of report.csv is the extension?
- Why can CSV open in Excel without becoming XLSX?
- Why is rename not conversion?

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE00-03

DE00-03 - L03 - Download, ZIP, Extract, Copy, Move and Rename

1. Why it matters

Projects often arrive as archives. Keeping source and working copies separate prevents accidental evidence loss.

2. What you are learning

Build and explain the competency: L03 - Download, ZIP, Extract, Copy, Move and Rename.

3. Prerequisite / starting state

Prerequisite: DE00-02 mastered with saved evidence.

4. Picture it in your head

A ZIP is a sealed travel bag. Extraction unpacks normal working copies; the ZIP can remain as the clean source.

5. New words before we use them

A ZIP is a sealed travel bag. Extraction unpacks normal working copies; the ZIP can remain as the clean source.

6. Useful video / source

Source: Microsoft Support — Zip and unzip files

Assigned scope: Create ZIP, Extract All, move/rename extracted files

How to use it: Follow the official Windows steps once, then perform the course ZIP/extract/copy/move/rename task without the guide.

Note: Current Windows 10/11 support page.

[Open source](#)

7. Follow with me once

Copy DOWNLOAD_ME_FIRST.zip to Downloads, Extract All, rename the extracted folder

m00_download_lab and move that working folder into

Documents\DE_COURSE_PRACTICE\01_MODULE_WORK\M00. Keep the original ZIP. Copy

DOWNLOAD_ME_FIRST.zip to Downloads, Extract All, rename the extracted folder m00_download_lab and move that working folder into Documents\DE_COURSE_PRACTICE\01_MODULE_WORK\M00. Keep the original ZIP.

8. Expected result / what you should see

Both the source ZIP and a separate editable extracted working folder exist.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Editing inside the ZIP preview.
- Moving the only clean source when a copy was intended.
- Downloading duplicate copies instead of locating the first one.

11. If you get stuck

Stop at the first point where your result diverges. Restate the intended input/state and expected output, inspect the smallest relevant step, reproduce only that step, and keep the failed attempt as evidence. Open protected review only after a genuine attempt has been saved.

12. Small independent task

Copy messy_downloads to messy_downloads_working and organize only the working copy. Keep source untouched. Evidence: Before/after folder listing proving the source is unchanged and the working copy was reorganized.

13. Validate

Both the source ZIP and a separate editable extracted working folder exist. Save a count, schema/key check, reconciliation, observed state, test, or other independent control appropriate to the lesson.

14. Teach it back

- What does extraction do?
- When choose Copy instead of Move?
- Why keep a clean source?

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE00-04

DE00-04 - L04 - Installing Software Safely

1. Why it matters

Blind early installation creates unnecessary failures and version conflicts.

2. What you are learning

Build and explain the competency: L04 - Installing Software Safely.

3. Prerequisite / starting state

Prerequisite: DE00-03 mastered with saved evidence.

4. Picture it in your head

An installer delivers/configures an application. PATH is a search list a shell uses to find command programs.

5. New words before we use them

An installer delivers/configures an application. PATH is a search list a shell uses to find command programs.

6. Useful video / source

Source: Microsoft Support — Install programs from online sources

Assigned scope: Trusted publisher/download/install workflow

How to use it: Use this as the safety overlay. Install only the tool requested by the current module and verify the publisher/source first.

Note: Official Microsoft support; course-specific install verification remains in the Teaching Book.

[Open source](#)

7. Follow with me once

Using VS Code as the example, identify the official download page, the installed Start-menu application and (if still present) the installer file. Do not install unrelated tools. Using VS Code as the example, identify the official download page, the installed Start-menu application and (if still present) the installer file. Do not install unrelated tools.

8. Expected result / what you should see

Learner can explain installer vs application, source trust, version verification and PATH without installing the whole DE stack.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Clicking sponsored download buttons.
- Approving UAC prompts without checking what launched them.
- Installing Python/PostgreSQL/WSL/Docker early "just in case".

11. If you get stuck

Stop at the first point where your result diverges. Restate the intended input/state and expected output, inspect the smallest relevant step, reproduce only that step, and keep the failed attempt as evidence. Open protected review only after a genuine attempt has been saved.

12. Small independent task

Pick one application already on the PC. Identify how you launch it and, if available, the old installer. Explain why they are different. Evidence: Short note with app name, launch location, installer location/status and verification method.

13. Validate

Learner can explain installer vs application, source trust, version verification and PATH without installing the whole DE stack. Save a count, schema/key check, reconciliation, observed state, test, or other independent control appropriate to the lesson.

14. Teach it back

- Installer vs installed app?
- Why use official source?
- What does PATH do?

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE00-05

DE00-05 - L05 - VS Code From Zero

1. Why it matters

Opening the correct root makes paths, terminals and later project structure understandable.

2. What you are learning

Build and explain the competency: L05 - VS Code From Zero.

3. Prerequisite / starting state

Prerequisite: DE00-04 mastered with saved evidence.

4. Picture it in your head

VS Code is a technical workspace/desk; it is not Python, SQL, Git or a database.

5. New words before we use them

VS Code is a technical workspace/desk; it is not Python, SQL, Git or a database.

6. Useful video / source

Source: Visual Studio Code — Official beginner tutorial

Assigned scope: 0:00–7:00

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

Note: Official VS Code visual orientation; the Teaching Book defines the exact course workspace.

[Open source](#)

7. Follow with me once

Open m00_download_lab as a folder, inspect data and notes, create evidence.txt at project root, save, close and reopen. Open m00_download_lab as a folder, inspect data and notes, create evidence.txt at project root, save, close and reopen.

8. Expected result / what you should see

Explorer shows the whole project tree and evidence.txt survives in the intended location.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Opening one random file instead of the project root.
- Creating a file in the wrong selected subfolder.
- Closing an Untitled/unsaved tab.

11. If you get stuck

Stop at the first point where your result diverges. Restate the intended input/state and expected output, inspect the smallest relevant step, reproduce only that step, and keep the failed attempt as evidence. Open protected review only after a genuine attempt has been saved.

12. Small independent task

Open supplier_import_assets as project root. Create evidence/hello.txt at the correct level, close and reopen, then verify location. Evidence: Project-tree screenshot or written path to evidence/hello.txt plus one sentence explaining project root.

13. Validate

Explorer shows the whole project tree and evidence.txt survives in the intended location. Save a count, schema/key check, reconciliation, observed state, test, or other independent control appropriate to the lesson.

14. Teach it back

- What is VS Code for?
- What is project root?
- Explorer vs Editor?

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE00-06

DE00-06 - L06 - Terminal Fundamentals

1. Why it matters

Many later DE tools are command-driven; location awareness prevents most beginner terminal mistakes.

2. What you are learning

Build and explain the competency: L06 - Terminal Fundamentals.

3. Prerequisite / starting state

Prerequisite: DE00-05 mastered with saved evidence.

4. Picture it in your head

File Explorer uses clicks; a terminal uses typed instructions. Every command runs from a current directory.

5. New words before we use them

File Explorer uses clicks; a terminal uses typed instructions. Every command runs from a current directory.

6. Useful video / source

Source: VS Code terminal orientation — Tech With Tim

Assigned scope: 4:40–6:40

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

Note: Short visual orientation only; the course lab teaches terminal safety and exact commands.

[Open source](#)

7. Follow with me once

From the VS Code terminal at m00_download_lab: pwd, ls, cd data, pwd, ls, cd .., mkdir terminal_created. From the VS Code terminal at m00_download_lab: pwd, ls, cd data, pwd, ls, cd .., mkdir terminal_created.

8. Expected result / what you should see

Learner can predict where the new folder appears and can explain each path change.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Running the right command from the wrong directory.
- Treating a path as if it were a command.
- Copying destructive commands without understanding them.

11. If you get stuck

Stop at the first point where your result diverges. Restate the intended input/state and expected output, inspect the smallest relevant step, reproduce only that step, and keep the failed attempt as evidence. Open protected review only after a genuine attempt has been saved.

12. Small independent task

From supplier_import_assets root, enter data, list files, return to root, create archive, enter it, create 2026, and record pwd at each level. Evidence: Saved terminal transcript or screenshot sequence showing the path changes and created folders.

13. Validate

Learner can predict where the new folder appears and can explain each path change. Save a count, schema/key check, reconciliation, observed state, test, or other independent control appropriate to the lesson.

14. Teach it back

- What is current directory?
- In mkdir evidence, command vs argument?
- What does cd .. do?

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE00-07

DE00-07 - L07 - Errors and Beginner Troubleshooting

1. Why it matters

Random multi-change troubleshooting hides the real cause and makes the next failure harder.

2. What you are learning

Build and explain the competency: L07 - Errors and Beginner Troubleshooting.

3. Prerequisite / starting state

Prerequisite: DE00-06 mastered with saved evidence.

4. Picture it in your head

An error is evidence. Symptom, cause and fix are different things.

5. New words before we use them

An error is evidence. Symptom, cause and fix are different things.

6. Useful video / source

Source: Teaching RC2 visual teaching + hands-on lab

How to use it: Use the Teaching Book and guided lab. No generic video is forced here; mastery still requires the independent task.

Note: This lesson is better learned from the Teaching Book + hands-on demonstration than from a generic external video.

Disposition: COURSE_BUILT / BOOK_ONLY for this lesson; no filler link is required.

7. Follow with me once

Use exercises/broken_setup/TASK.txt. Read the symptom, find the misplaced file, repair only the location problem and write repair_note.txt while preserving broken_setup_source. Use exercises/broken_setup/TASK.txt. Read the symptom, find the misplaced file, repair only the location problem and write repair_note.txt while preserving broken_setup_source.

8. Expected result / what you should see

One controlled fix with a repair note that names symptom, cause, change and verification.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Reinstalling software to fix a path problem.
- Changing several settings at once.
- Deleting the broken original evidence.

11. If you get stuck

Stop at the first point where your result diverges. Restate the intended input/state and expected output, inspect the smallest relevant step, reproduce only that step, and keep the failed attempt as evidence. Open protected review only after a genuine attempt has been saved.

12. Small independent task

Use exercises/path_problem/TASK.txt. Save attempt note first, form one hypothesis at a time, fix the path issue and write a two-sentence cause/repair explanation. Evidence: attempt_note.txt plus repaired state plus explanation of cause and verification.

13. Validate

One controlled fix with a repair note that names symptom, cause, change and verification. Save a count, schema/key check, reconciliation, observed state, test, or other independent control appropriate to the lesson.

14. Teach it back

- Why change one thing at a time?
- For file not found, what inspect first?
- What makes a repair note useful?

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE00-08

DE00-08 - L08 - Study Evidence, Attempts and Review Discipline

1. Why it matters

Job work and interviews require recall and reasoning without an answer sheet.

2. What you are learning

Build and explain the competency: L08 - Study Evidence, Attempts and Review Discipline.

3. Prerequisite / starting state

Prerequisite: DE00-07 mastered with saved evidence.

4. Picture it in your head

Recognition is not recall. Preserve the first attempt because it shows what you actually knew before review.

5. New words before we use them

Recognition is not recall. Preserve the first attempt because it shows what you actually knew before review.

6. Useful video / source

Source: Teaching RC2 visual teaching + hands-on lab

How to use it: Use the Teaching Book and guided lab. No generic video is forced here; mastery still requires the independent task.

Note: This lesson is better learned from the Teaching Book + hands-on demonstration than from a generic external video.

Disposition: COURSE_BUILT / BOOK_ONLY for this lesson; no filler link is required.

7. Follow with me once

Create 02_ATTEMPTS, 03_REVIEW, 04_RETESTS, 05_EVIDENCE. Save m00_test_attempt.txt, then a separate m00_test_retest.txt and evidence note. Create 02_ATTEMPTS, 03_REVIEW, 04_RETESTS, 05_EVIDENCE. Save m00_test_attempt.txt, then a separate m00_test_retest.txt and evidence note.

8. Expected result / what you should see

Original attempt remains unchanged; retest is separate; evidence identifies both.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

- Editing the first attempt after seeing review.
- Calling Reviewed the same as Mastered.
- Calling Skipped the same as Mastered.

11. If you get stuck

Stop at the first point where your result diverges. Restate the intended input/state and expected output, inspect the smallest relevant step, reproduce only that step, and keep the failed attempt as evidence. Open protected review only after a genuine attempt has been saved.

12. Small independent task

Create attempts/P05_attempt1.txt with an intentionally wrong sentence, then retests/P05_retest1.txt with correction; explain both in evidence/P05_evidence.txt. Evidence: Three separate files proving attempt, retest and explanation.

M00 move-on standard Move to M01 only when you can find the course workspace, identify common file formats, extract and preserve source archives, explain safe installation, open a project root in VS Code, navigate a terminal safely, troubleshoot one path problem with one hypothesis at a time, and preserve attempts/retests/evidence separately.

13. Validate

Original attempt remains unchanged; retest is separate; evidence identifies both. Save a count, schema/key check, reconciliation, observed state, test, or other independent control appropriate to the lesson.

14. Teach it back

- Why save first attempt?
- Review vs retest?
- Why is skipped not mastered?

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: Proceed to the module practical/Gate route after the module mastery checklist.

Module mastery checklist + revision quick reference

- ☐ Every lesson has a saved genuine independent attempt.
- ☐ Every lesson has at least one independent validation/control appropriate to the tool.
- ☐ Protected reviews were opened only after attempts; any weak lesson has a changed retry after review was closed.
- ☐ You can teach back each lesson in your own words and name one realistic failure mode.
- ☐ Required module practical/retest evidence is saved and no critical defect is unresolved.
- ☐ D+1, D+4, D+10 and D+30 revision events are scheduled for weak or high-value competencies.

Revision cadence

D+1: fresh micro-task. D+4: mixed retrieval. D+10: changed/failure case. D+30: interview/project retrieval. Always attempt from memory before reopening a source.

Release boundary

This book is a static learner artifact. It does not prove live tool/cloud execution or a protected Gate pass. Runtime evidence must be recorded separately when the curriculum requires it.